

Analyzing the importance of recycling

Bachelor Summer Project (Academic Year 2025/2026), University of Luxembourg

Ene Alexandra
alexandra.ene.001@student.uni.lu
Université du Luxembourg

Samuel Desire Kom Sande
samuel.kom@uni.lu
Université du Luxembourg

Martin Theobald
martin.theobald@uni.lu
Université du Luxembourg

Abstract

Recycling is an important component of sustainable waste management, but the effectiveness of recycling system depends not only on the availability of recycling technologies, but also on the behavior of the individuals. Ways of raising awareness have become more of a popular topic, as the problem with waste management increases and the health of our planet decreases.

To bring this problem to an individual level, the way household waste is separated before collection can influence the quality of materials available for recycling. Recent research [1] shows that plastics separated at the household level generally contain fewer contaminants than plastics recovered later from mixed residual waste, highlighting the continued importance of source separation in recycling systems.

This project investigates how a simple educational game can have an impact on the exposure and reinforcement of basic recycling rules. We developed a two dimensional game using the LOVE2D framework and the Lua programming language. The core idea is the different types of waste fall from the top of the screen, and the player must move the recycling bin and select the correct waste color category before catching each item. Correct classification will increase the player's score while incorrect one decreases it. The game is separated into two modes: easy, where there are only 3 waste types, and normal, with 5 types. The difficulty progressively increases as the player's score grows, and the game is completed when the player reaches 50 points.

The project tries to make education interactive and entertaining, but also accessible to people with a lower exposure to technology, that may not understand very well a more complex program.

1 Introduction

Waste management has become an environmental challenge because large quantities of materials are consumed and discarded every year. Plastic waste is particularly problematic. According to recent research, [2] global plastic production has exceeded 400 million tones per year, while only a relatively small proportion is recycled.

One important stage occurs before waste even reaches a recycling facility: household waste separation. Incorrect sorting can introduce contamination into recyclable waste streams and can reduce the quality of the material recovered.

The article Good recycling starts at home-and benefits the world [3], highlights the importance of sorting waste correctly at home. It explains that plastics separated by households are generally cleaner and less contaminated than plastics recovered later from mixed waste, which can improve the quality of materials available for recycling.

Industrial sorting is still important because it can recover recyclable materials that were not separated at home. However, it works best as a complement to household sorting rather than a replacement for it. Using both methods together can make recycling more effective

Poor waste management can harm the environment in several ways. Landfills can pollute groundwater and release methane, while throwing away recyclable materials also increases the need to extract new resources.

At the European level, increasing recycling is consequently part of the transition towards a circular economy. European waste policy establishes recycling and collection objectives for several waste streams. Nevertheless, recycling rates remain below their full potential, showing that infrastructure and policy targets alone are not enough[4].

Human behavior is also essential for effective recycling. People need clear and simple rules to understand where each type of waste belongs. When recycling systems are easy and convenient to use, people are more likely to participate, while complicated systems can discourage them.

This project explores one possible educational approach: a small interactive recycling game. Rather than explaining recycling only through written instructions, the game asks players to actively classify waste while responding to increasing time pressure. Repetition, immediate feedback and progressively increasing difficulty are used to transform basic recycling information into an interactive task.

2 Problem Statement

The recycling system mostly depends on individuals being able to separate correctly household waste. A person that has access to the recycling bins and still makes mistakes will still cause a problem.

The problem addressed in this project can therefore be expressed as follows: How can a simple educational game help users practice and understand basic household waste sorting categories?

The project focuses especially on familiarizing players with the association between different waste types and their corresponding bins. The complete version of the game contains five categories: Mixed, Organic, Glass, Paper and Plastic.

These categories are represented using different bin colours so that users must learn both the waste category and its visual association.

A second challenge is keeping players engaged. An informational screen stating recycling rules might communicate the information, but it provides little interaction. Therefore, the project uses a game based approach in which the player continuously makes classification decisions.

The game also incorporates gradual difficulty. At the beginning, waste appears less frequently and can fall relatively slowly. As the

player's score increases, the spawning interval becomes shorter and the possible falling speed becomes higher. The difficulty changes at scores 10, 20, 30 and 40. This encourages the player to become faster at recognizing the correct categories rather than completing the game through a fixed repetitive rhythm.

The final objective is reached at a score of 50. A successful player must repeatedly demonstrate correct sorting decisions while adapting to increasing speed.

The project is designed primarily as an educational prototype. It does not attempt to represent every recycling rule or every possible household object. Instead, it investigates whether a compact interactive game can provide an understandable introduction to waste sorting and whether users consider this approach useful.

3 Game Design

The recycling game was developed in Lua using the LOVE2D game framework. It uses a simple two dimensional design so that the main interaction is immediately understandable and the recycling task remains the central focus.

3.1 Core Gameplay

The player controls a single recycling bin located at the bottom of the screen. Waste objects are generated at random horizontal positions above the screen and fall vertically towards the bottom.

The player has two responsibilities: move the bin horizontally so that it is positioned underneath the falling object and change the type of the bin so that it matches the falling waste, as seen in Figure 1.

The bin can be moved using the left and right arrow keys or the A and D keys.

The recycling category is selected using number keys. In Normal Mode, the controls are: 1-Mixed, 2-Organic, 3-Glass, 4-Paper, 5-Plastic.

Each waste category and its corresponding bin use the same colour. This creates a direct visual relationship between the falling object and the appropriate bin.

When an object collides with the bin, the program compares the identifier of the falling waste with the currently selected bin type.

If the two categories match, the score increases by 1, otherwise it decreases by 1.

3.2 Game Modes

Two game modes were implemented to provide different levels of complexity.

3.2.1 Easy Mode. Easy Mode-Figure 3- contains three categories: Mixed, Plastic and Glass.

Only these three types of waste are generated during the game.

This mode reduces the number of possible choices and can therefore be used by players who are unfamiliar with the system or who want to learn the basic mechanics before moving to the complete version.

3.2.2 Normal Mode. Normal Mode contains all five waste categories: Mixed, Organic, Glass, Paper and Plastic.



Figure 1: Game on normal difficulty

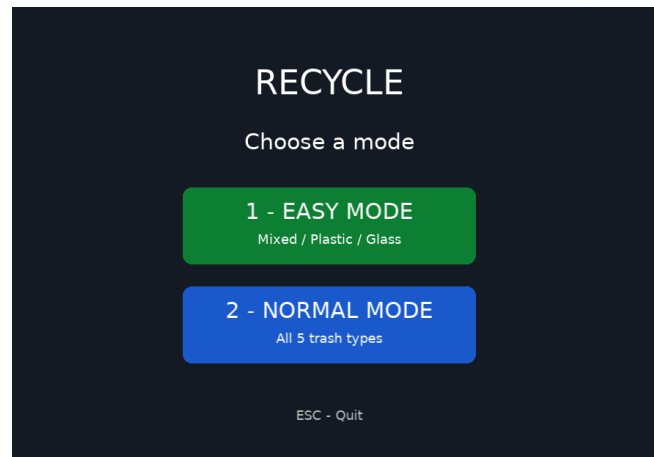


Figure 2: Game menu

Normal Mode-Figure 1-requires the player to distinguish between more categories and therefore increases the cognitive difficulty of the classification task.

Before gameplay starts, the program displays a menu in which the user chooses between Easy Mode and Normal Mode, as shown in Figure 2.

3.3 Random Waste Generation

Waste objects are generated randomly. The program randomly chooses one of the waste categories available in the selected game mode and assigns the object a random horizontal position.

The speed is also selected from a range rather than being identical for every object. As a consequence, players cannot simply memorize a constant timing pattern.

The combination of random waste category, horizontal position and falling speed makes each game session slightly different.

3.4 Progressive Difficulty

The difficulty increases according to the player's score.



Figure 3: Game on easy difficulty

Score	Spawn interval	Falling-speed range
0–9	2.0 s	80–240
10–19	1.6 s	100–270
20–29	1.3 s	120–300
30–39	1.1 s	140–330
40–49	0.9 s	160–360
50	Game completed	–

Table 1: Game difficulty progression based on score.

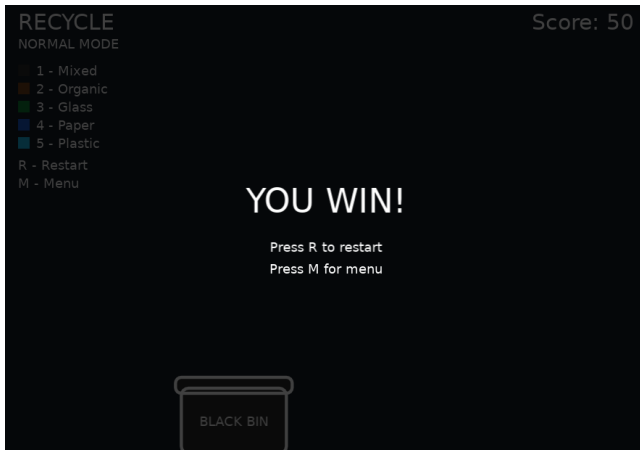


Figure 4: Win screen

At score 10, objects begin appearing more frequently and the maximum falling speed increases. The process repeats at scores 20, 30 and 40.

3.5 Win, Restart and Menu Systems

The game is completed when the player's score reaches 50. At this point, falling objects are removed and "YOU WIN!" screen is displayed, as shown in Figure 4.

The restart and menu controls are also displayed during game-play.

The menu is used to select between the 2 game difficulties.

4 Evaluation/Experiments

The next stage of the project should evaluate how users interact with the game and whether they perceive it as useful for learning recycling categories.

4.1 Demographic Questions

The demographic section of the questionnaire was used to obtain basic information about the participants and their recycling habits. A total of 10 participants took part in the study. The questions collected information about age, gender and the frequency with which participants recycle household waste.

As shown in Table 2, the participants reported different levels of recycling frequency. Three participants stated that they rarely recycle, three reported recycling sometimes, and another three reported recycling often. One participant indicated that they always recycle, while none of the participants selected "Never". Overall, the results show that all participants had at least some previous experience with household recycling, although the frequency varied considerably.

Table 3 presents the gender distribution of the participants. Seven participants were female and three were male.

The age distribution is presented in Table 4. Participants ranged in age from 8 to 51 years, providing responses from individuals belonging to different age groups. Most ages were represented by one participant, with age 19 being the only age reported by two participants.

These demographic results provide context for interpreting the questionnaire responses collected after playing the game. In particular, the variation in age and recycling frequency allows the evaluation to include participants with different levels of previous exposure to household recycling practices

Table 2: Household Recycling Frequency

Response	Participants
Never	0
Rarely	3
Sometimes	3
Often	3
Always	1

Table 3: Gender Distribution

Response	Participants
Female	7
Male	3

4.2 Questions After Playing

After playing the game, participants completed a questionnaire evaluating the usability, educational value, engagement, and perceived

Table 4: Age Distribution

Age	Participants
8	1
17	1
19	2
20	1
21	1
24	1
40	1
46	1
51	1

difficulty of the application. The results are summarized in Table 5. For most questions, responses were measured on a five point scale, where 1 represented "Strongly disagree" and 5 represented "Strongly agree".

The game was generally considered easy to understand. Question 1 obtained a mean score of 4.0, with eight out of ten participants selecting either 4 or 5. The controls were also evaluated positively, although responses were slightly more varied, resulting in a mean score of 3.6.

The use of colors to distinguish between the recycling categories received a mean score of 4.0. This suggests that the color based design helped most participants identify the appropriate recycling bin. Similarly, the question asking whether the game helped participants remember which type of waste belongs to which bin received a mean score of 4.2, one of the highest scores in the questionnaire.

Participants also generally considered the game useful for learning about recycling. The 5th question received a mean score of 3.6. The increasing difficulty of the game received a mean score of 3.5, indicating that participants generally found this feature engaging.

Question 7 measured the perceived difficulty of the game using a different scale, where 1 represented "Very easy" and 5 represented "Very difficult". The mean score was 2.7. Five participants selected a difficulty level of 2, three selected 3, and two selected 4. This indicates that the game was generally perceived as being between easy and moderately difficult.

The highest mean score, 4.4, was obtained for the question asking whether games can be an effective way to teach children or teenagers about recycling. Nine out of ten participants selected either 4 or 5, suggesting a strong positive perception of game based approaches for recycling education.

Finally, participants were asked how confident they felt about correctly identifying the recycling categories used in the game after playing. This question obtained a mean score of 3.8, with seven out of ten participants selecting either 4 or 5. Overall, the questionnaire results indicate that participants generally found the game understandable, useful, and suitable as an educational tool for practicing basic recycling categories.

4.3 Further Feedback

One suggestion was to add sound effects, particularly when the player loses points or makes an incorrect classification. This could provide clearer immediate feedback and make the game more engaging.

Sound effects could also be used for positive actions, such as correctly catching a waste item or reaching a new difficulty level. However, these effects should remain simple so that they support the learning experience without distracting the player.

This feedback suggests that future improvements should not only focus on gameplay mechanics and visual design, but also on audio feedback as a way of making player actions and consequences easier to understand.

5 Future Work

The current project represents an initial prototype, and several features could be developed further.

One important improvement would be replacing the coloured rectangular waste objects with images or sprites representing real household items. For example, the game could display bottles, newspapers, food packaging, jars and food waste instead of showing only category labels. This would make the classification task more realistic because users would need to recognize an object rather than simply match a written category.

A second improvement would involve introducing more detailed recycling rules. Real household recycling is more complex than five broad categories. Packaging can contain several materials, some objects require special collection systems, and recycling instructions can differ between municipalities. The current simplified design could therefore be expanded gradually after the basic categories have been learned.

Another possible feature would be a short explanation after an incorrect answer. Instead of only subtracting one point, the game could display a message such as: Incorrect, glass belongs in the green bin.

Another direction would be localization. Recycling systems vary between countries, meaning that the game could load different categories, colours and instructions depending on the player's location.

Furthermore, another idea was to integrate artificial intelligence, particularly large language models, into the game. If a player places a waste item in the wrong recycling bin, the AI could provide a short and personalized explanation describing why the choice was incorrect and indicating the correct recycling category.

This could make mistakes more useful from an educational perspective, since the player would receive immediate feedback rather than simply being informed that the answer was wrong. Over time, such explanations could help users better understand the reasoning behind recycling rules and reduce repeated mistakes. The use of an LLM could also allow the feedback to be adapted to the user's level of understanding, providing simpler explanations for younger players and more detailed information for users who want to learn more.

6 Conclusion

This project investigated the use of a simple educational game to introduce household recycling categories. The motivation for the project comes from the importance of correct waste separation at the household level

The developed LOVE2D game converts this idea into an interactive activity. Players must move a recycling bin, change its category

Table 5: Questionnaire Results After Playing the Game

Questions	1	2	3	4	5	Mean
1 Was it easy to understand how the game works?	0	0	2	6	2	4.0
2 Were the controls easy to use?	0	1	4	3	2	3.6
3 Did the colours make it easier to identify the correct bin?	0	1	1	5	3	4.0
4 Did the game help you remember which waste belongs to which bin?	0	0	1	6	3	4.2
5 Did you learn something about recycling while playing?	0	1	3	5	1	3.6
6 Did the increasing difficulty make the game more engaging?	0	1	4	4	1	3.5
7 How difficult was the game overall?	0	5	3	2	0	2.7
8 Do you think games can be an effective way to teach children or teenagers about recycling?	0	0	1	4	5	4.4
9 After playing the game, how confident are you that you can correctly identify the recycling categories used in the game?	0	0	3	6	1	3.8

and correctly catch falling waste. Two game modes allow the complexity to be adjusted: Easy Mode and Normal.

The game also introduces progressive difficulty. Objects appear increasingly frequently and can fall faster as the player reaches scores of 10, 20, 30 and 40. Reaching 50 points completes the game.

Although the application is intentionally simple, it demonstrates how basic environmental information can be transformed into an interactive learning task.

Ultimately, recycling depends on both technological systems and human participation. Making recycling information clearer and easier to practice is therefore one small but relevant contribution to improving household waste separation.

7 References

- Mathias Cormann, *OECD. Global Plastics Outlook: Economic Drivers, Environmental Impacts and Policy Options. OECD* https://www.oecd.org/en/publications/global-plastics-outlook_de747aef-en/full-report.html
- Alexandra Schmuck, Tiago G. A. Belé, Daniël Withoeck, Kevin M. Van Geem, Kim Ragaert, Steven De Meester, *Analysis of trade-offs of post-sorting plastic packaging. Nature* <https://www.nature.com/articles/s41586-026-10606-4>
- Editorial, *Good recycling starts at home — and benefits the world* <https://www.nature.com/articles/d41586-026-01823-y>
- European Environment Agency, *Reaching 2030's residual municipal waste target — why recycling is not enough* <https://www.eea.europa.eu/en/analysis/publications/reaching-2030s-residual-municipal-waste-target-why-recycling-is-not-enough>

